

DISCLAIMER: Exam Warning

For the exam, assume ALL material covered in the lectures is relevant!

⇒ The exam can include problems similar to problems uploaded, but not worked through in the lectures!

BEL 2 20260610

□ Building the diagonalization from logistics modes.

A 2-hub distribution process has 2 independent right eigenvectors

$$v_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix} \quad v_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix},$$

with corresponding eigenvalues

$$\lambda_1 = 1, \quad \lambda_2 = 0.4$$

a) Explain in one logistics sentence what it means that v_1 is preserved in direction under the redistribution matrix.

⇒ Answer: The 1:2 stock distribution ratio is an unchanging equilibrium state for this logistics network.

Explanation: Mathematically, an eigenvector is one that preserves its direction when multiplied by a matrix.

$$A v_1 = \lambda_1 v_1$$

$$\text{Our case} \quad \Rightarrow \quad 1 \cdot \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

Barring disruptions, the trucks moving stock will maintain the 1:2 stock ratio.

b) Construct the matrix P , whose columns are the eigenvectors v_1 and v_2 .

$\Rightarrow$ Change of Basis P :

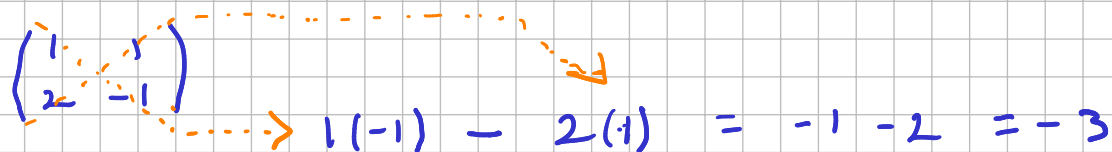
$$P = (v_1 \ v_2) = \begin{pmatrix} 1 & 1 \\ 2 & -1 \end{pmatrix}$$

c) Construct the diagonal matrix D , whose diagonal entries are the corresponding eigenvalues.

$$\Rightarrow D = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0.4 \end{pmatrix}$$

d) Compute $\det(P)$; explain why $\det(P) \neq 0$ is necessary for diagonalization.

$\det(P)$:


$$\begin{pmatrix} 1 & 1 \\ 2 & -1 \end{pmatrix} \rightarrow 1(-1) - 2(1) = -1 - 2 = -3$$

$$\det(P) = -3$$

Explanation: $\det(P) \neq 0$ is a necessary condition for diagonalization because

$$A = PDP^{-1}$$

implies the inverse of P , which is P^{-1} .
A matrix is only invertible if its determinant is non-zero.

e) Compute P^{-1}

$$P = \begin{pmatrix} 1 & 1 \\ 2 & -1 \end{pmatrix}, \quad \det(P) = -3$$

$$P^{-1} = \frac{1}{\det(P)} \operatorname{adj}(P)$$

Given matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$

1. Swap diagonal elements a & d .
2. Change the signs of the off-diagonal elements b & c .
3. Multiply by $\frac{1}{\det(P)}$

$$\Rightarrow P = \begin{pmatrix} 1 & 1 \\ 2 & -1 \end{pmatrix}, \quad \det(P) = -3$$

$$1. \quad \begin{pmatrix} -1 & 1 \\ 2 & 1 \end{pmatrix}$$

$$2. \quad \begin{pmatrix} -1 & -1 \\ -2 & 1 \end{pmatrix} = \operatorname{adj}(P)$$

$$3. \quad \frac{1}{-3} \begin{pmatrix} -1 & -1 \\ -2 & 1 \end{pmatrix} = \begin{bmatrix} \frac{1}{3} & \frac{1}{3} \\ \frac{2}{3} & -\frac{1}{3} \end{bmatrix}$$

2 Constructing a redistribution matrix from PDP^{-1}

Given

$$P = \begin{pmatrix} 1 & 1 \\ 2 & -1 \end{pmatrix}, \quad D = \begin{pmatrix} 1 & 0 \\ 0 & 0.4 \end{pmatrix}.$$

a) Compute $A = PDP^{-1}$

$$\Rightarrow \text{We know } P^{-1} = \begin{pmatrix} 1/3 & 1/3 \\ 2/3 & -1/3 \end{pmatrix}$$

$\Rightarrow$ First compute PD , then multiply by P^{-1}

$$PD = \begin{pmatrix} 1 & 1 \\ 2 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 0.4 \end{pmatrix} = \begin{pmatrix} 1(1) + 0 & 0 + 1(0.4) \\ 2(1) + 0 & 0 + -1(0.4) \end{pmatrix}$$

$$PD = \begin{pmatrix} 1 & 0.4 \\ 2 & -0.4 \end{pmatrix} \checkmark$$

$$PDP^{-1} = (PD)P^{-1}$$

$$= \begin{pmatrix} 1 & 0.4 \\ 2 & -0.4 \end{pmatrix} \begin{pmatrix} 1/3 & 1/3 \\ 2/3 & -1/3 \end{pmatrix} = \begin{pmatrix} 1(1/3) + 0.4(2/3) & 1(1/3) + 0.4(-1/3) \\ 2(1/3) + (-0.4)(2/3) & 2(1/3) + (-0.4)(-1/3) \end{pmatrix}$$

$$PDP^{-1} = \begin{pmatrix} 0.6 & 0.2 \\ 0.4 & 0.8 \end{pmatrix}$$

b) Verify directly that:

$$Av_1 = 1 \cdot v_1 \quad \Rightarrow \quad Av_1 = \lambda_1 v_1$$

$$\Rightarrow \text{We know: } A = PDP^{-1} = \begin{pmatrix} 0.6 & 0.2 \\ 0.4 & 0.8 \end{pmatrix}$$

$$Av_1 = \begin{pmatrix} 0.6 & 0.2 \\ 0.4 & 0.8 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 0.6(1) + 0.2(2) \\ 0.4(1) + 0.8(2) \end{pmatrix} \\ = \begin{pmatrix} 1 \\ 2 \end{pmatrix} \quad \checkmark = 1 \cdot \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

$$Av_2 = \begin{pmatrix} 0.6 & 0.2 \\ 0.4 & 0.8 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 0.6(1) + 0.2(-1) \\ 0.4(1) + 0.8(-1) \end{pmatrix} \\ Av_2 = \begin{pmatrix} 0.4 \\ -0.4 \end{pmatrix} = 0.4 \begin{pmatrix} 1 \\ -1 \end{pmatrix} \quad \checkmark$$

$\Rightarrow$ Interpretation: This shows a total conservation of stock. All stock at source hubs is guaranteed to arrive at the destination hubs.

d) Identify the dominant eigenvalue and the dominant eigenvector. What long-run hub ratio do they imply?

$\Rightarrow$ The dominant eigenvalue is λ_1 because it has the largest absolute value

$$|\lambda_1| = 1. \quad \checkmark$$

The dominant eigenvector is therefore v_1 . This implies the long-run hub ratio is 1:2.

* The eigenvalue $|\lambda_2| = 0.4$ corresponds to the transient mode.